Bearings

Year 4 · **Lesson Worksheet**

Worked Examples

Three-figure bearings: Write the bearing for a direction that is 35° measured clockwise from North, as a three-figure bearing.

Answer: A bearing is always written using three digits, so 35° is written as 035°.

Compass to bearing: What is the approximate three-figure bearing for South-East (SE)?

Answer: SE lies exactly halfway between South (180°) and East (090°), so its bearing is 135°.

Back bearings: A hiker walks from point A to point B on a bearing of 072°. Find the bearing of A from B (the back bearing).

Answer: Since 072° is less than 180°, add 180°: 072° + 180° = 252°. The back bearing of A from B is 252°.

Bearings Basics (15 questions)

1. What is the bearing of North? Give your answer as a three-figure bearing.
2. What is the bearing of East? Give your answer as a three-figure bearing.
3. Write 8° as a three-figure bearing.
4. Write 240° as a three-figure bearing. A) 24 B) 024 C) 240 D) 0240 E) 24.0
5. A ship sails on a bearing of 090°. In which compass direction is it travelling?
6. What is the bearing of South, as a three-figure bearing?
7. A plane flies on a bearing of 180°. In which compass direction is it travelling? A) North B) East C) South D) West E) North-East
8. List the 8 compass directions in clockwise order starting from North.
9. What is the approximate bearing of North-East (NE)?
10. A walker is told to head on a bearing of 315°. Which compass direction does this match? A) North-East B) South-East C) South-West D) North-West E) West
11. Select ALL that apply: which of the following are valid three-figure bearings? A) 045° B) 7° C) 199° D) 099° E) 450°
12. What is the bearing of West, as a three-figure bearing?
13. A hiker walks on a bearing of 090° and then turns to walk on a bearing of 270°. What is the relationship between these two directions?
14. Select ALL that apply: which of these bearings point in a generally southerly direction (between 090° and 270°)? A) 060° B) 150° C) 200° D) 300° E) 350°
15. A compass needle points on a bearing of 000°. Which compass direction is this? A) North B) East C) South D) West E) North-East

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Reading & Finding Bearings (15 questions)

1. A ship's radar shows a lighthouse at a bearing of 050° from the ship. Describe, using a compass direction, roughly where the lighthouse lies relative to the ship.
2. The bearing of town B from town A is 065°. What is the back bearing (the bearing of A from B)?
3. The bearing of a hill from a campsite is 142°. What is the back bearing of the campsite from the hill?
4. A bearing of 300° is given. What is the back bearing? A) 100° B) 110° C) 120° D) 130° E) 140°
5. A boat sails from harbour P to harbour Q on a bearing of 210°. What is the bearing from Q back to P?
6. Select ALL that apply: which of these pairs correctly show a bearing and its exact back bearing? A) 020° and 200° B) 095° and 185° C) 160° and 340° D) 250° and 070° E) 300° and 110°
7. A bearing of 075° is measured from North, turning clockwise. Through how many degrees would you turn, starting by facing North, to face along this bearing?
8. Two signposts show that Hilltown is on a bearing of 128° from Riverside. What direction, roughly (to the nearest compass point of N, NE, E, SE, S, SW, W, NW), is Hilltown from Riverside? A) East B) South-East C) South D) South-West E) West
9. A bearing of 190° is given. What is the back bearing?
10. A plane flies on a bearing of 048° from airport X to airport Y. Describe the back bearing in words (using a compass direction) and give its value as a three-figure bearing.
11. A bearing of 358° is given. What is the back bearing? A) 088° B) 178° C) 180° D) 182° E) 268°
12. Select ALL that apply: a ship travels from a buoy to a jetty on a bearing of 085°. Which statements about the return journey (jetty to buoy) are true? A) The back bearing is 265°. B) The back bearing is found by subtracting 180° from 085°. C) The back bearing is found by adding 180° to 085°. D) The return journey is on a bearing of less than 180°. E) 085° and 265° differ by exactly 180°.
13. A bearing from a ship to a buoy is measured as 212°. What is the bearing from the buoy back to the ship?
14. On a map, a church is due North-West of a school. Give the bearing of the church from the school.
15. A diagram shows two points, C and D. The bearing of D from C is 310°. State the bearing of C from D.

Bearings Challenge (15 questions)

1. A hiker walks from camp to a stream on a bearing of 060°. What bearing must she walk on to get back from the stream to the camp?
2. A delivery van drives from the depot to a shop on a bearing of 072°. What bearing does it use to drive straight back to the depot?
3. Flying from city P to city Q uses a bearing of 128°. On the return flight from Q to P, what bearing is needed?
4. From one starting point, a drone flies off on a bearing of 045° while a second drone flies on a bearing of 165°. How big is the angle between their flight paths? A) 100° B) 110° C) 120° D) 130°
5. A cyclist rides on a bearing of 100°, then turns 90° clockwise. What bearing is she now cycling on?
6. Select ALL that apply: the bearing from a house to a park is 155°. Which of these statements are true? A) The back bearing (park to house) is 335°. B) The park lies roughly to the south of the house. C) The back bearing is found by subtracting 180° from 155°. D) The house lies roughly to the north of the park. E) The angle between the bearing and its back bearing is 180°.
7. When point X was surveyed, its bearing to point Y was recorded as 210°, but the instrument was faulty and under-read by 10°. What is the real bearing from X to Y, and what back bearing does that give?

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8. Two ships leave the same harbour at the same time: Ship 1 on a bearing of 020°, Ship 2 on a bearing of 310°. What is the angle between their two courses? A) 60° B) 70° C) 80° D) 90°

9. A plane flies from airport A on a bearing of 085° to airport B, then from airport B on a bearing of 265° to airport C. What does this tell you about point C? A) C is the same point as A. B) C is further from A than B is. C) C is due north of A. D) There is not enough information to say.

10. From a lighthouse, a passing boat is spotted on a bearing of 300°. If the crew on the boat looked back at the lighthouse, what bearing would they measure for it?

11. Select ALL that apply: a hiking bearing from point M to point N is 064°. Which of the following are correct ways to describe the return bearing from N to M? A) 064° + 180° = 244° B) 244° C) 180° − 064° = 116° D) The return bearing and the outward bearing differ by 180°. E) 064° − 180° = −116°, so add 360° to get 244°.

12. A ship sails away from a harbour on a bearing of 210°. It then turns around and sails straight back towards the harbour. What bearing is it now sailing on?

13. Suppose a bearing is 128°. After converting it to a back bearing, which one of the 8 main compass points is it nearest to? A) South B) North C) South-West D) North-West E) North-East

14. A ship sails on a bearing of 080° for 10 km to point B, then on a bearing of 260° for 4 km to point C. How far is C from the ship's original starting point?

15. The bearing from a cabin to a lake is 070°. The back bearing from the lake to the cabin is 70° plus some number of degrees. What number must be added to 70° to get the back bearing?

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